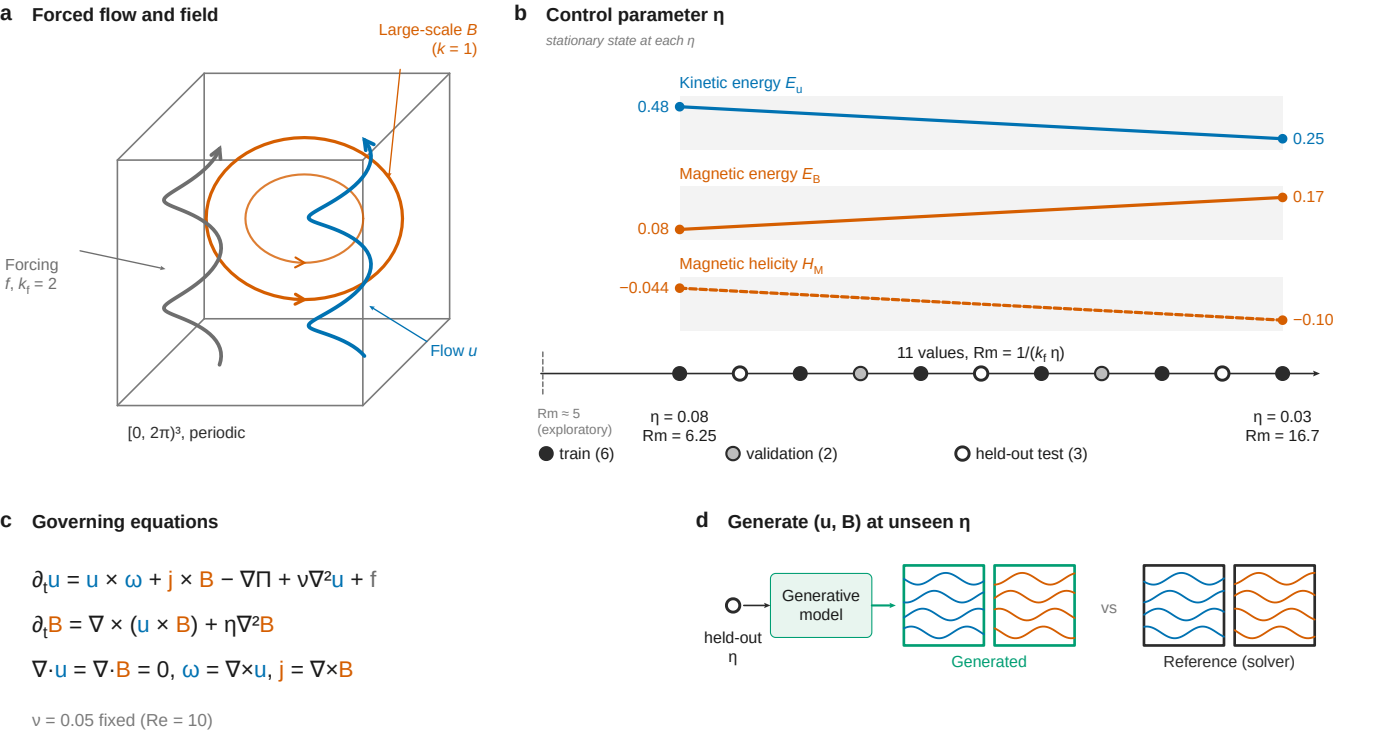

a
Forced flow and field
Large-scale B
(k = 1)
Forcing
f, k_f = 2
Flow u
[0, 2π)³, periodic
b
Control parameter η
stationary state at each η
Kinetic energy E_u
0.48
0.25
Magnetic energy E_B
0.08
0.17
Magnetic helicity H_M
−0.044
−0.10
11 values, Rm = 1/(k_f η)
Rm ≈ 5
(exploratory)
η = 0.08
Rm = 6.25
η = 0.03
Rm = 16.7
train (6)
validation (2)
held-out test (3)
c
Governing equations
∂_t u = u × ω + j × B − ∇Π + ν∇²u + f
∂_t B = ∇ × (u × B) + η∇²B
∇·u = ∇·B = 0, ω = ∇×u, j = ∇×B
ν = 0.05 fixed (Re = 10)
d
Generate (u, B) at unseen η
held-out
η
Generative
model
Generated
vs
Reference (solver)